

1. Write a program to accept a number and check if it is a Mark number or not. A number is said to be Mark when the sum of the squares of each digit is an even number as well as the last digit of the sum and the last digit of the number given are same.

Example: $n = 246$

$$\text{sum} = 2 \times 2 + 4 \times 4 + 6 \times 6 = 56$$

2. A tech number has even number of digits. If the number is split in two halves, then the square of sum of these halves is equal to the number itself. Write a program to generate and print all four-digit tech numbers.

Example:

Consider the number 3025

$$\text{Square of sum of the halves of } 3025 = (30 + 25)^2$$

$$= (55)^2$$

$$= 3025 \text{ is a tech number.}$$

3. Write a program to input a number and check and print whether it is a Pronic number or not. Pronic number is the number which is the product of two consecutive integers.

Examples:

$$12 = 3 \times 4$$

$$20 = 4 \times 5$$

$$42 = 6 \times 7$$

4. Define a class to accept a number from user and check if it is an EvenPal number or not. The number is said to be EvenPal number when number is palindrome number (a number is palindrome if it is equal to its reverse) and sum of its digits is an even number.

Example: 121 is a palindrome number.

Sum of the digits = $1 + 2 + 1 = 4$ which is even number.

5. Write a menu-driven program using switch case to accept a choice from the user and according to the choice entered perform the following operations:

(a) To calculate and display the area of a circle using the formula:

$$\text{Area} = (\pi \times \text{radius}^2) \text{ where } \pi = 22 / 7$$

(b) To calculate and display the area of a rectangle using the formula:

$$\text{Area} = (\text{length} \times \text{breadth})$$

(c) To calculate and display the area of a triangle using the formula:

$$\text{Area} = 1 / 2 \times \text{base} \times \text{height}$$

(d) To calculate and display the area of a square using the formula:

$$\text{Area} = \text{side}^2$$

For an incorrect option, an appropriate error message should be displayed.

56 is an even number as well as last digit is 6 for both sum as well as the number.

6. Define a class to overload the method format() as follows:

void format(): To print the following pattern using nested for loops only:

1 2 3 4 5

2 3 4 5

3 4 5

4 5

5

int format(String s): To calculate and return the sum of ASCII codes of each character of the string.

Example: CAB

Output: $67 + 65 + 66 = 198$

void format(int n): To calculate and display the sum of natural numbers up to n given by the formula $n(n + 1) / 2$

7. Design a class to overload a function series() as follows:

(a) void series(int x, int n) - to display the sum of the series given below:

$x_1 + x_2 + x_3 + \dots + x_n$ terms

(b) void series(int p) - to display the following series:

0, 7, 26, 63, ... p terms

(c) void series() - to display the sum of the series given below:

$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{10}$

8. Design a class to overload a function volume() as follows:

(i) double volume(double r) - with radius (r) as an argument, returns the volume of sphere using the formula: $v = \frac{4}{3} \times \frac{22}{7} \times r^3$

(ii) double volume(double h, double r) - with height (h) and radius (r) as the arguments, returns the volume of a cylinder using the formula: $v = \frac{22}{7} \times r^2 \times h$

(iii) double volume(double l, double b, double h) - with length(l), breadth (b) and height (h) as the arguments, returns the volume of a cuboid using the formula: $v = l \times b \times h$

9. Define a class named StepTracker with the following specifications:

Member Variables:

String name: stores the user's name

int sw: stores the total number of steps walked by the user

double cb: stores the estimated calories burned by the user

double km: stores the estimated distance walked in kilometers

Member Methods:

void accept(): to input the name and the steps walked using Scanner class methods only

void calculate(): calculates calories burned and distance in km based on steps walked using the following estimation table:

Metric	Calculation Formula
Calories Burned	steps walked * 0.04 (e.g. 1 step burns 0.04 calories)
Distance (km)	steps walked / 1300 (e.g. 1300 steps is approx. 1 km)

void display(): Display the calories burned, distance in km and the user's name.

Write a main() method to create an object of the class and invoke the methods.

10. Design a class ShowRoom with the following description:

Instance variables/data members:

String name - to store the name of the customer

long mobNo - to store the mobile number of the customer

double cost - to store the cost of the items purchased

double dis - to store the discount amount

double amount - to store the amount to be paid after discount

Member methods:

ShowRoom() - default constructor to initialize data members

void input() - to input customer name, mobile number, cost

void calculate() - to calculate discount on the cost of purchased items, based on the following criteria:

Cost	Discount (in percentage)
Less than or equal to ₹ 10000	5%
More than ₹ 10000 and less than or equal to ₹ 20000	10%
More than ₹ 20000 and less than or equal to ₹ 35000	15%
More than ₹ 35000	20%

void display() - to display customer name, mobile number, amount to be paid after discount
Write a main() method to create an object of the class and call the above member methods.

11. Define a class to initialize the following data in an array.

Search for a given character input by the user, using the Binary Search technique.

Print "Search successful" if the character is found otherwise print "Search is not successful".

'A', 'H', 'N', 'P', 'S', 'U', 'W', 'Y', 'Z', 'b', 'd'

12. Define a class pincode and store the given pin codes in a single dimensional array. Sort these pin codes in ascending order using the selection sort technique only. Display the sorted array.

110061, 110001, 110029, 110023, 110055, 110006, 110019, 110033

13. Define a class to accept 10 characters from a user. Using bubble sort technique to arrange them in ascending order. Display the sorted array and original array.

14. Define a class to accept values into an array of double data type of size 20. Accept a double value from user and search in the array using linear search method. If value is found display message "Found" with its position where it is present in the array. Otherwise display message "not found".

15. Write a program to accept name and total marks of N number of students in two subscript array name[] and totalMarks[].

Calculate and print:

(i) The average of the total marks obtained by N number of students. [average = (sum of total marks of all the students) / N]

(ii) Deviation of each student's total marks with the average. [deviation = total marks of a student - average]

16. Write a program to accept a two-dimensional integer array of order 4×5 as input from the user.

Check if it is a Sparse Matrix or not. A matrix is considered to be a sparse, if the total number of zero elements is greater than the total number of non-zero elements. Print appropriate messages.

Example:

```
4 3 0 1 0
1 0 0 2 0
1 0 1 0 0
0 3 2 0 0
```

Number of zero elements = 11

Number of non-zero elements = 9

Matrix is a Sparse Matrix

17. Write a program to input and store integer elements in a double dimensional array of size 3×3 and find the sum of elements in the left diagonal.

Example:

```
1 3 5
4 6 8
9 2 4
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Output:

Sum of the left diagonal elements = $(1 + 6 + 4) = 11$

18. Write a program to accept a word and print the Symbolic of the accepted word. Symbolic word is formed by extracting the characters from the first consonant, then add characters before the first consonant of the accepted word and end with "TR".
Example: AIRWAYS Symbolic word is RWAYSITR
BEAUTY Symbolic word is BEAUTYTR
19. Write a program to input a sentence and convert it into uppercase and count and display the total number of words starting with a letter 'A'.
Example:
Sample Input: ADVANCEMENT AND APPLICATION OF INFORMATION TECHNOLOGY ARE EVER CHANGING.
Sample Output: Total number of words starting with letter 'A' = 4.
20. Write a program in Java to accept a string in lowercase and change the first letter of every word to uppercase. Display the new string.
Sample input: we are in cyber world
Sample Output: We Are In Cyber World
21. Accept a string and form an array with each of its character using a library function. Print the characters in reverse order .
22. WAP in java to accept (input)the names and total marks of girls opting for arts from class V111A . Also accept the names and total marks of girls opting for arts from V111B .Merge the two arrays to form the resultant array of 1XC .
Print: a)Each of the three arrays of names with proper message
b)Sort the array of 1XC (ascending order)depending upon their total marks (Use SELECTION SORT technique) and display the output in the following manner:
- | | |
|-------|--------|
| NAME: | MARKS: |
| NAME: | MARKS: |
23. WAP to enter 20 real numbers in a single dimensional array and find the following:
- Print the sum of the alternate elements
 - Product of the even elements
 - Average of the elements less than 20.0
 - 2nd largest element along with its position
 - Rounded up values of each element down to its nearest integer.
24. WAP to pass a character array as an argument and perform the following:
- Arrange the characters in descending order and print them
 - Check if characters form a palindrome or not
 - Print the uppercase characters in a row.
 - Print the no of lowercase characters
 - No of spaces
 - No. of digits or special characters.
 - No. of times 'b' occurs in the array.
25. Create the function Bill and display the final amount to be paid by each of the n customers and other details At an Online shopping App .
- Check for proceeding (Input customer's name, product-name, product-code, amount to be paid and shipping address of each customer).
 - Way / Mode of Payment(C-Credit Card, D-Debit Card, N-Net Banking, P-Pay on Delivery).
 - Ask whether the user continues to shop or not .